



THE STATE UNIVERSITY OF ZANZIBAR
SCHOOL OF COMPUTING COMMUNICATION AND MEDIA STUDIES
DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION
TECHNOLOGY
FINAL EXAMINATION 2024/2025
SEMESTER II
DATA MINING AND BUSINESS INTELLIGENCE - INF 3202

Date: 16/07/2025

Time: 09:00 AM - 12:00 AM

INSTRUCTIONS

1. This paper consists of **TWO** sections, **A** and **B**, which carries 24 and 36 marks respectively.
2. Answer all questions from section A, and any **THREE** questions from section B.
3. Cellular phones and any other unauthorized materials are **NOT** allowed in the examination room.
4. **ANSWER EACH QUESTION ON SEPERATE SHEET**
5. This exam consists of eight (7) printed pages, including cover page

SECTION A (24 marks)

Answer ALL questions from this section.

Question 1: For each item, choose the correct answer and write down its corresponding letter (1 mark each)

- (i) Which of the following best describes a data warehouse and its key characteristics?
- A. Transactional database used for routine operations and real-time updates
 - B. A system for storing daily logs with minimal redundancy and real-time writes.
 - C. A subject-oriented, integrated, time-variant, and non-volatile collection of data for decision making.
 - D. A decentralized storage system used to increase transactional throughput.
- (ii) What is the primary function of Online Analytical Processing (OLAP)?
- A. To manage large-scale transactional updates in real-time databases.
 - B. To process unstructured social media data.
 - C. To perform multidimensional analysis of data for complex queries and decision support.
 - D. To back up and restore enterprise databases.
- (iii) Which of the following is *not* typically a goal of a data warehouse in modern organizations?
- A. To provide historical data for analysis and reporting.
 - B. To support complex queries for decision making.
 - C. To handle day-to-day transactional updates.
 - D. To integrate data from multiple sources for unified analysis.
- (iv) Which of the following are types of clusters used in clustering analysis?
- A. Centroid-based and density-based clusters
 - B. Hierarchical and linear clusters
 - C. Temporal and spatial clusters only
 - D. Numeric and symbolic clusters
- (v) Which of the following best describes a data mining algorithm?
- A. A method used to clean duplicate records in a database.
 - B. A machine learning model used to process raw input into audio signals.

- C. A computational method for discovering patterns in large datasets; such as decision trees for classification.
 - D. A function that controls access to cloud resources.
- (vi) What is the main purpose of Business Intelligence (BI) in organizations?
- A. To automate office workflows and document management.
 - B. To support data-driven decision making through analysis, visualization, and reporting.
 - C. To control hardware operations and device drivers.
 - D. To reduce manufacturing costs through automation.
- (vii) Which of the following groups represents the three primary disciplines that support data mining technology?
- A. Statistics, Machine Learning, Database Systems
 - B. Networking, Programming, Hardware Engineering
 - C. Web Development, System Administration, Graphic Design
 - D. Artificial Intelligence, Cybersecurity, Software Development
 - E. Statistics, Software Development, Database Systems
- (viii) Which layer in a typical data mining architecture handles user queries and reporting?
- A. Data source layer
 - B. Data warehouse layer
 - C. Presentation layer
 - D. ETL layer
- (ix) Which of the following is a common algorithm used in classification tasks?
- A. K-Means
 - B. Decision Tree
 - C. Apriori
 - D. DBSCAN
- (x) What is the main goal of a data mart?
- A. To store real-time sensor data
 - B. To support operational processes
 - C. To serve a specific business function or department
 - D. To replace the entire enterprise database

- (xi) Which characteristic is NOT typically associated with a data warehouse?
- A. Volatile data updates
 - B. Integrated data
 - C. Subject-oriented data
 - D. Time-variant data
- (xii) Which of the following SQL queries is most appropriate for extracting summarized sales data (total sales per region) from a fact table in a data warehouse?
- A. `SELECT * FROM sales_fact;`
 - B. `SELECT region, COUNT(*) FROM sales_fact;`
 - C. `SELECT region, SUM(sales_amount) FROM sales_fact GROUP BY region;`
 - D. `SELECT region, sales_amount FROM sales_fact WHERE sales_amount > 1000;`
- (xiii) Which of the following best describes a standalone data mining system?
- A. A data mining tool that is fully integrated with a data warehouse system
 - B. A web-based application for visualizing dashboards and reports
 - C. An independent data mining system that performs data processing and analysis without relying on external data warehouses
 - D. A cloud-based storage service that supports distributed data mining
- (xiv) Which of the following correctly distinguishes between supervised and unsupervised learning?
- A. Supervised learning uses input data without labels, while unsupervised learning requires labeled data.
 - B. Supervised learning groups data into clusters, while unsupervised learning predicts future outcomes.
 - C. Supervised learning uses labeled data to train models for prediction, while unsupervised learning identifies hidden patterns in unlabeled data.
 - D. Both supervised and unsupervised learning always require predefined classes.

(xv) In a star schema used for data warehousing, what is the primary difference between fact tables and dimension tables?

- A. Fact tables store descriptive attributes, while dimension tables store numerical measures.
- B. Fact tables contain transactional or measurable data, while dimension tables provide context through descriptive attributes.
- C. Fact tables are used only during ETL, while dimension tables are used for backups.
- D. Fact tables are temporary, whereas dimension tables are permanent.

Question 2

(a) Data mining architecture is composed of multiple interconnected components. Fill in the missing details in Figure 1 to complete the diagram [3 marks].

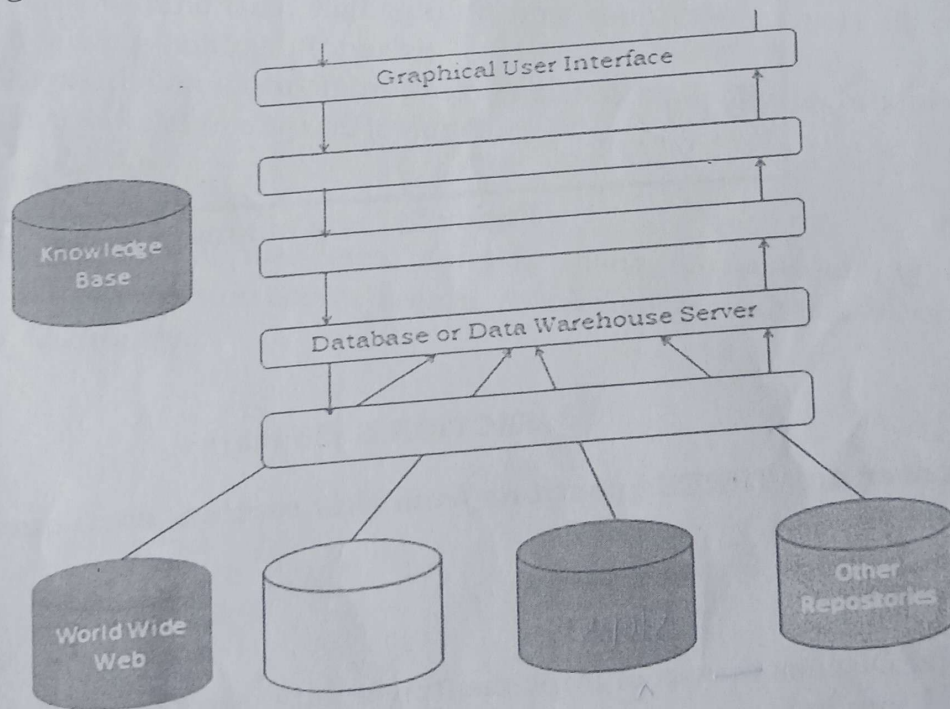


Figure 1: Data mining architecture

- (b) (i) Complete the Figure 2 by accurately labeling all the key steps involved in the data mining process [3 marks].
- (ii) For each step you have identified, briefly describe its purpose in one or two sentences [3 marks].

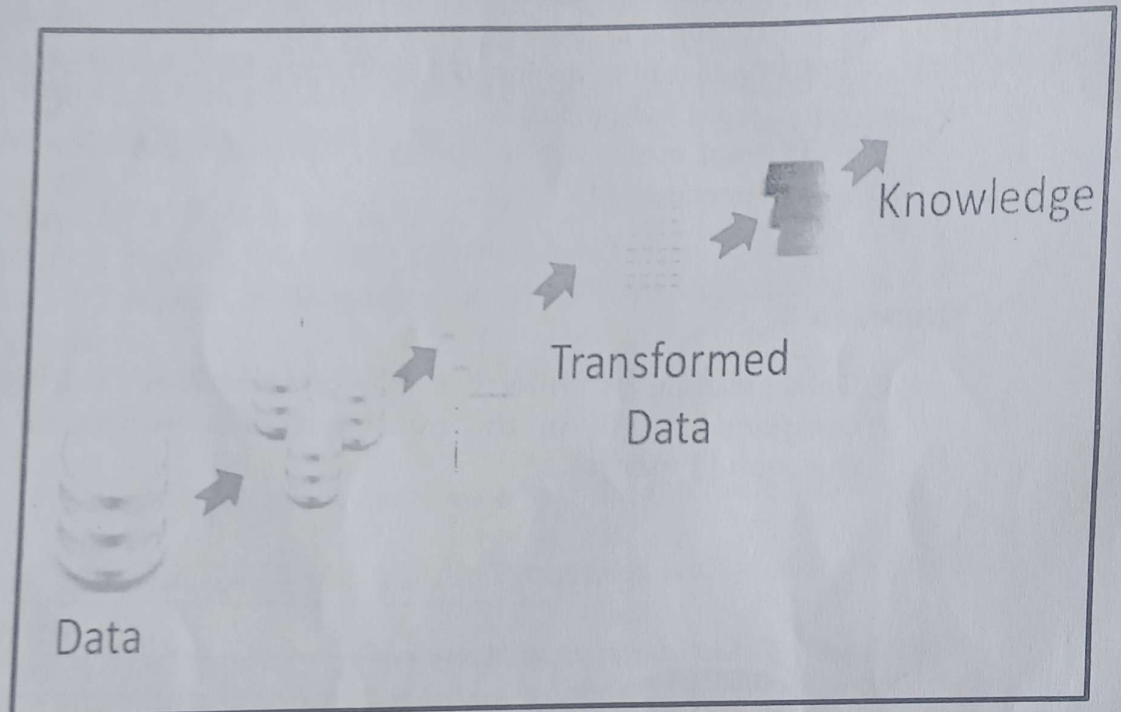


Figure 2: Data mining process

SECTION B (36 marks)

Answer any THREE questions from this section, each question carries 12 marks.

Question 3

Steps

- (a) Discuss ~~five~~ ^{six} of implementing a data warehouse in an enterprise. [6 marks]
- (b) Explain how data mining algorithms can be applied to predict supply chain risks in the company. [6 marks]

Question 4

Design a simple star schema for a retail electronic store's data warehouse. Identify at least one fact table and three dimension tables, explaining the role of each table and how they relate to one another in enabling multidimensional analysis. [12 marks]

Question 5

- (a) Describe the clustering technique in data mining and provide one example of how clustering can be used in market segmentation or anomaly detection. [8 marks]
- (b) Briefly explain the importance of data preparation in data mining and describe any two key activities involved in the process. [4 marks]

Question 6

- (a) Critically evaluate the differences between operational databases and data warehouses, highlighting how each can support different types of decision-making. [8 marks]
- (b) Explain how data mining dimensional modeling techniques differ from Entity-Relationship (ER) modeling. [4 marks]

Question 7

Discuss the ethical and legal considerations of data mining in business intelligence applications with regard to predictive analytics in organizational settings. [12 marks]